GW

Mann, Michael <mmann1123@email.gwu.edu>

Decision on submission to Computers and Electronics in Agriculture

Computers and Electronics in Agriculture <em@editorialmanager.com> Fri, Oct 2, 2026 at 4:08 AM
Reply-To: Computers and Electronics in Agriculture <support@elsevier.com>
To: Michael Mann <mmann1123@gwu.edu>

Ref.: Ms. No. **COMPAG-D-26-07034**
Sparse Feature Selection, Not Deep Learning Capacity, Drives Spatial Transfer in Crop Classification
Computers and Electronics in Agriculture

Dear Professor Mann,
Thank you for submitting your manuscript to Computers and Electronics in Agriculture. I have completed the review of your manuscript . The reviewers recommend reconsideration of your paper following major revision. I invite you to resubmit your manuscript after addressing all reviewer comments.

To view your reviewer feedback, please log in as an author at https://www.editorialmanager.com/compag/ and navigate to your manuscript in the " Submissions Needing Revision " folder under the Author Main Menu.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully. Please note that your revised submission may need to be re-reviewed.

Please resubmit your manuscript by **Nov 01, 2026**.

Research Elements (optional)
This journal encourages you to share research objects - including your raw data, methods, protocols, software, hardware and more – which support your original research article in a Research Elements journal. Research Elements are open access, multidisciplinary, peer-reviewed journals which make the objects associated with your research more discoverable, trustworthy and promote replicability and reproducibility. As open access journals, there may be an Article Publishing Charge if your paper is accepted for publication. Find out more about the Research Elements journals at https://www.elsevier.com/authors/tools-and-resources/research-elements-journals?dgcid=ec_em_research_elements_email.

I look forward to receiving your revised manuscript.

Yours sincerely
Fernando Auat
Associate Editor
Computers and Electronics in Agriculture
Comments from the Editors and Reviewers:

Reviewer's Responses to Questions

Note: In order to effectively convey your recommendations for improvement to the author(s), and help editors make well-informed and efficient decisions, we ask you to answer the following specific questions about the manuscript and provide additional suggestions where appropriate.

1. Are the objectives and the rationale of the study clearly stated?

Please provide suggestions to the author(s) on how to improve the clarity of the objectives and rationale of the study. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: Yes

Reviewer #2: The experimental theme of this manuscript is to analyze the accuracy performance of models under a pure transfer setting without any target-region labels. However, the title uses “Spatial Transfer.” It is recommended to revise the title so that it is more consistent with the basic experimental setting of the paper and avoids overstating the spatial transfer capability.

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Please provide suggestions to the author(s) on how to improve the replicability/reproducibility of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes
Provide further comments here:

Reviewer #2: The authors classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. The authors need to show the phenological characteristics of these five crop types. Since several of these crops may be easily confused, the authors also need to explain whether these crop types can be accurately identified using only time-series features

---

3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Please clearly indicate if the manuscript requires additional peer review by a statistician. Kindly provide suggestions to the author(s) on how to improve the statistical analyses, controls, sampling mechanism, or statistical reporting. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes
Provide further comments here:

Reviewer #2: The main model used in this paper is L-TAE-S, which adds a sparse-attention encoder, while most of the comparison models are basic machine-learning and deep-learning models, including LightGBM, XGBoost, CNN-BiLSTM, and TabNet. It is recommended to add recent crop time-series classification models for a more objective comparison and analysis.

The experiment only performs transfer classification between adjacent regions. The training regions are 34S_19E_258N and 34S_19E_259N, while the holdout region is 34S_20E_259N. The phenological characteristics, field shapes, and natural environmental conditions of the training and transfer regions are almost the same. Therefore, this experimental setting cannot be used to fully judge the spatial transfer performance of the model.

---

4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Please provide specific suggestions for improvements, removals, or additions of figures or tables. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: Yes

Reviewer #2: The authors need to add figures showing the spatial distributions of crop classification results from L-TAE-S and the other comparison models, and further analyze the classification accuracy of the models from the perspective of spatial distribution.

---

5. If applicable, are the interpretation of results and study conclusions supported by the data?

Please provide suggestions (if needed) to the author(s) on how to improve, tone down, or expand the study interpretations/conclusions. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Mark as appropriate with an X:
Yes
Provide further comments here:

Reviewer #2: After spatial transfer, the accuracy improvement of the proposed model compared with the other models is limited. The transfer macro-F1 is 0.60 for L-TAE-S, 0.60 for TabNet, 0.56 for XGBoost, and 0.58 for the original L-TAE. With this level of accuracy, the model is still difficult to apply to operational crop mapping. The authors need to further clarify the practical application of the proposed method.

---

6. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Please provide suggestions to the author(s) on how to better emphasize the strengths of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: Yes

Reviewer #2: yes

---

7. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Please list the limitations that the author(s) need to add or emphasize. Please number each limitation so that author(s) can more easily respond.

Reviewer #1: Yes

Reviewer #2: It is recommended to further discuss the crop classification accuracy of L-TAE-S under different transfer years, different transfer-area sizes, and different transfer distances.

---

8. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Please provide suggestions to the author(s) on how to improve the manuscript structure and flow. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: No

Reviewer #2: no

---

9. Could the manuscript benefit from language editing?

Reviewer #1: Yes

Reviewer #2: Yes

Reviewers have commented and despite the suggestion of further analysis, comments can be addressed in due time and novelty is not questioned. Therefore the decision. When proceeding with the revised version, please also prepare a very detailed response letter addressing each of the comments provided. Thank you. Additionally, we would expect authors to perform a world-wide search for preparing the state of the art review and only use those articles that are related to the research. Fernando Auat Cheein, associate editor.

Reviewer #1: This paper focuses on the core challenge of model spatial generalization in agricultural remote-sensing crop classification. Using the Sentinel-2 time-series dataset for the Western Cape, South Africa, it systematically evaluates the performance differences among 23 models spanning classical machine learning, deep learning, and patch-based models. The study identifies sparse feature selection—rather than deep-learning capacity—as the key determinant of spatial transfer performance, and proposes the L-TAE-S sparse attention temporal encoder. The methodology of this work is innovative; however, several issues remain. Detailed comments are provided below.

1. The Abstract describes the benchmark as comprising "twenty-three classical and deep models," whereas the Introduction and Methodology explicitly distinguish three paradigms: classical machine learning, representation-learning deep networks, and patch-based spatial models. Since the patch-based models are also deep-learning-based architectures, the current terminology may cause ambiguity. Please clarify and consistently describe the three model categories throughout the manuscript.

2. Section 3.2.1 only compares L-TAE and L-TAE-S to verify the effect of the sparse gate, without performing sensitivity analysis for key parameters such as sparsity hyperparameters (γ, λ) and the number of attention heads. Accordingly, the sensitivity of the reported transfer performance to these hyperparameters remains unclear. Supplementary ablation experiments investigating the impacts of key hyperparameters on transfer performance are recommended.

3. The supplementary ablation analysis strengthens the manuscript by comparing L-TAE, Transformer, and L-TAE-S, yet it still does not provide sufficient evidence to substantiate the stronger claim that sparsity itself drives spatial transfer. Specifically, the reported ST macro-F1 improves only marginally from 0.58 (L-TAE/Transformer) to 0.60 (L-TAE-S), a gain of merely 0.02. Notably, the manuscript itself acknowledges that ST macro-F1 differences below approximately 0.05

should be regarded as statistically indistinguishable. Consequently, the observation that "L-TAE-S performs slightly better" is insufficient to support the conclusion that "sparsity is the mechanism that determines transfer."

4. The mathematical formulation of the sparsity penalty in L-TAE-S requires further clarification. The main text states that "a penalty λ∑_(h,t)‖ m_(h,t) ‖_1(λ=10^(-3)) adds mild additional pressure", while the supplementary materials note that "At each of nsteps decision steps, an attentive transformer produces a sparse feature mask via sparsemax, M[i] = sparsemax". If this term corresponds to the standard output of sparsemax, it lies within the probability simplex m_i≥0,∑_im_i =1. Accordingly, it would be a constant term given a fixed number of heads or time steps, and theoretically cannot drive additional sparsity. The authors need to explicitly elaborate the practical implementation of this regularization term and verify whether it genuinely participates in the optimization process. If the formulation is consistent with the code implementation, the functional role of this regularization term needs to be reinterpreted.

5. Section 4.5 only compares the performance of xr_fresh features and raw reflectance on XGBoost, and concludes that "xr_fresh has no accuracy advantage but only operational value". The authors should make this scope explicit wherever this finding is discussed and avoid implying that the same conclusion necessarily applies to deep temporal models. If feasible, an analogous comparison for one representative temporal model (e.g., L-TAE) would further clarify whether the observed representation effect is model-dependent.

6. The paper attributes the unsatisfactory performance of 2D/3D CNNs, the paper attributes it to "limited spatial context + lack of sparsity", while overlooking factors including patch size, receptive field configuration, and spatial temporal feature fusion strategies. It is suggested to add comparative experiments with varying patch sizes, or soften the causal interpretation and discuss these alternative factors.

7. The manuscript identifies small-grain grazing, barley, and wheat as the most difficult classes and reports that these categories account for a substantial proportion of the accuracy loss under spatial transfer. However, the study does not propose improvement ideas or analysis for this core difficulty. It is recommended to supplement discussion on classification strategies for spectrally similar crops, or verify the effect of feature enhancement and fine-grained classification methods on the transfer performance of these categories.

8. The study uses only adjacent tiles within the same growing season and agroecological zone as the holdout set, constituting a local spatial shift. The authors also acknowledge in the limitations section that this setup would underestimate the accuracy degradation associated with cross-regional or cross-year transfer. It is recommended that the authors either supplement the study with cross-year or cross-ecozone transfer experiments, or more clearly delineate the applicable boundaries of the conclusions in the discussion.

Reviewer #2: This field is optional. If you have any additional suggestions beyond those relevant to the questions above, please number and list them here.

Have questions or need assistance?
For further assistance, please visit our customer service site: http://help.elsevier.com/app/answers/list/p/9435/. Here you can search for solutions on a range of topics, find answers to frequently asked questions, and learn more about Editorial Manager via interactive tutorials. You can also talk 24/5 to our customer support team by phone and 24/7 by live chat and email.

At Elsevier, we want to help all our authors to stay safe when publishing. Please be aware of fraudulent messages requesting money in return for the publication of your paper. If you are publishing open access with Elsevier, bear in mind that we will never request payment before the paper has been accepted. We have prepared some guidelines (https://www.elsevier.com/connect/authors-update/seven-top-tips-on-stopping-apc-scams) that you may find helpful, including a short video on Identifying fake acceptance letters (https://www.youtube.com/watch?v=o5l8thD9XtE). Please remember that you can contact Elsevier s Researcher Support team (https://service.elsevier.com/app/home/supporthub/publishing/) at any time if you have questions about your manuscript, and you can log into Editorial Manager to check the status of your manuscript (https://service.elsevier.com/app/answers/detail/a_id/29155/c/10530/supporthub/publishing/kw/status/).

#AU_COMPAG#

To ensure this email reaches the intended recipient, please do not delete the above code

---

In compliance with data protection regulations, you may request that we remove your personal registration details at any time. (Remove my information/details). Please contact the publication office if you have any questions.